

3 Add a "print" and "input()"

The program doesn't show what it's doing at each step. Adding a "print" command here will let you see what's happening. The "input()" command waits for the "return" or "Enter" key to be pressed before looping.

```
top_num = 5
total = 0
for n in range(top_num):
    total = total + n
    print('DEBUG: n=', n, 'total=', total)
    input()
print('Sum of numbers 1 to', top_num, 'is', total)
```

This command prints the current value of the loop variable and the total so far



4 New output

The loop is only adding the numbers from 0 up to 4, and not 1 to 5. This is because a "for" loop always starts counting from 0 (unless told otherwise), and always stops 1 before the end of the range.

```
DEBUG: n= 0 total= 0
DEBUG: n= 1 total= 1
DEBUG: n= 2 total= 3
DEBUG: n= 3 total= 6
DEBUG: n= 4 total= 10
Sum of numbers 1 to 5 is 10
```

This is actually the sum of the numbers from 0 to 4, not 1 to 5



5 Fix the faulty line

The range should go from 1 up to "top_num + 1", so that the loop adds up the numbers from 1 to "top_num" (5).

```
top_num = 5
total = 0
for n in range(1, top_num + 1):
    total = total + n
    print('DEBUG: n=', n, 'total=', total)
    input()
print('Sum of numbers 1 to', top_num, 'is', total)
```

The new range will count from 1 and stop at "top_num" (1 less than "top_num + 1")



6 Correct output

The "print" command shows that the program is adding the numbers from 1 to 5 and getting the correct answer. The bug has now been fixed!

```
DEBUG: n= 1 total= 1
DEBUG: n= 2 total= 3
DEBUG: n= 3 total= 6
DEBUG: n= 4 total= 10
DEBUG: n= 5 total= 15
Sum of numbers 1 to 5 is 15
```

When "n= 3", the total is (1 + 2 + 3)

The correct answer is now printed

